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Interview Kit: PostgreSQL Query Performance & Indexing

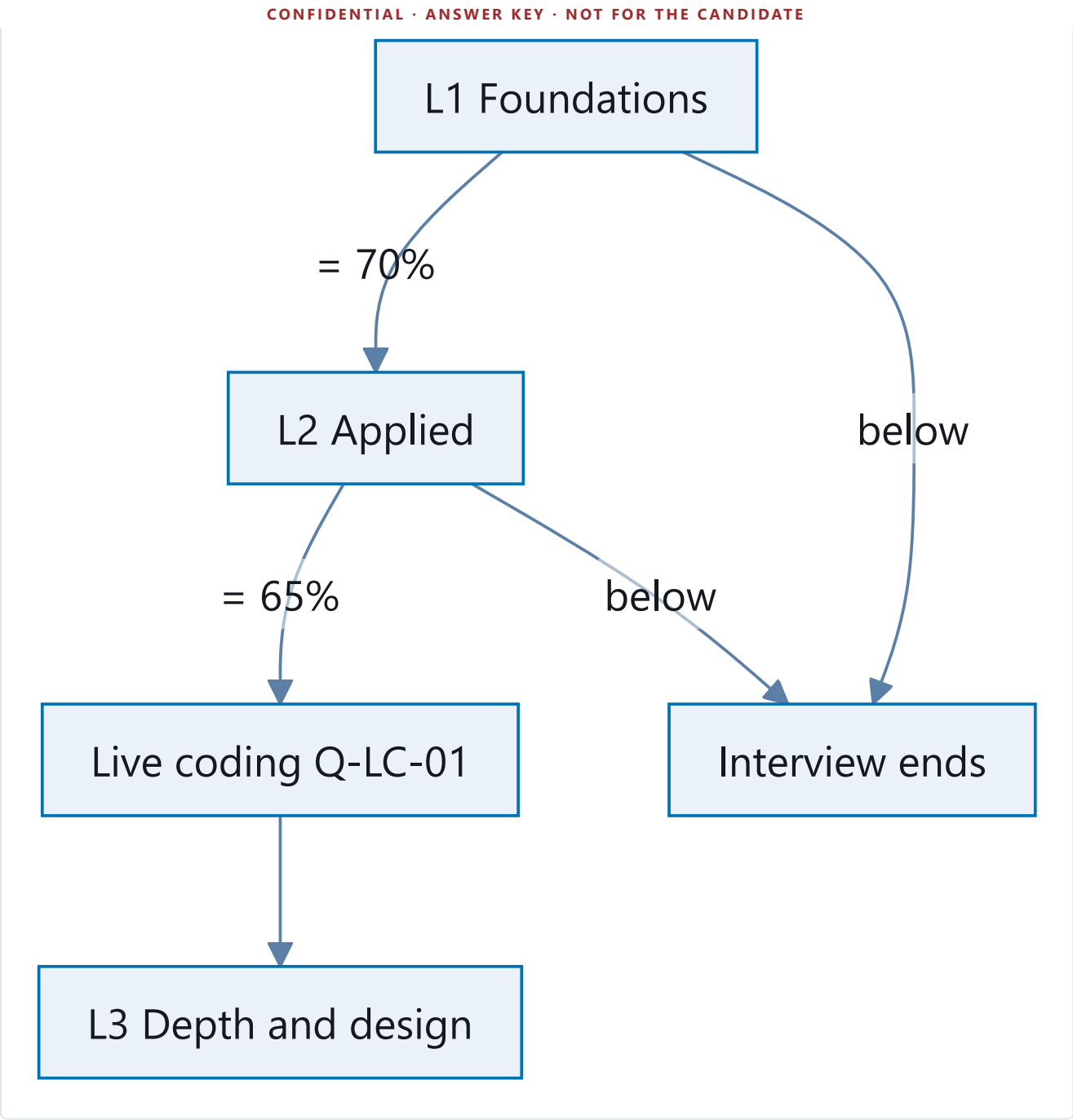
KIT.MD | INTERVIEWER | 2026-09-02

- **Kit ID:** postgres-indexing
- **Built:** 2026-08-30
- **Source file:** example/source-postgres-indexing.md
- **Role target:** Mid-level Backend Engineer (inferred from source — "new backend engineers", 200ms p99 budget, no schema-design ownership implied)
- **Total questions:** 6, including one live-coding task (abbreviated demonstration kit; a real kit is 15–20)
- **Question files:** questions.md
- **Contract version:** 1

Levels

Level	Name	Questions	Pass threshold	Gate
L1	Foundations	2	70%	Must pass to reach L2
L2	Applied	2 + Q-LC-01	65%	Must pass to reach L3
L3	Depth & Design	1	60%	Terminal

Levels run in order and the interview stops at the first gate that is not cleared:



Live coding

- **Segment:** yes
- **Placement:** after L2's questions, scored into L2
- **Questions:** Q-LC-01
- **Total budget:** 20 min
- **Mode:** review-only
- **Runtime:** PostgreSQL 16 — no live database is provisioned for this demonstration kit, so the query is scored by reading it.
- **Environment prerequisites:** none. A `runnable` version of this kit would require a seeded PostgreSQL 16 instance and would ship `livecode/Q-LC-01/tests.sql`; `livecode-contract.md` describes that path, which this abbreviated kit does not demonstrate.

Scope

IN SCOPE B-tree structure and ordering; composite index column order; causes of index non-use; reading EXPLAIN ANALYZE; the write cost of indexes; selectivity and planner decisions; writing an index-aware query under a latency budget.

OUT OF SCOPE replication, partitioning strategy, connection pooling (excluded by the source); anything requiring access to Acme's real schema.

Research provenance

> This demonstration kit was built **without web research** — no source was fetched. Every row below is therefore labelled `model knowledge`, which is what the builder must do when web tools are unavailable. A real kit fetches the primary documentation and cites it with a retrieval date.

Claim / area	Source	Retrieved	Confidence
Function-wrapped predicates prevent B-tree index use absent a matching expression index	model knowledge	n/a	high
Composite index usable for a leading prefix of its columns, not an arbitrary subset	model knowledge	n/a	high
Each index adds write amplification; HOT updates skip index maintenance when no indexed column changes and the page has room	model knowledge	n/a	med
Planner prefers a seq scan above a selectivity threshold; no fixed universal ratio	model knowledge	n/a	med

Known-unstable areas

- **Planner cost defaults** — `random_page_cost` defaults and their practical effect differ across versions and on SSD-tuned installs. Accept any answer that names the setting and reasons about it correctly; do not require a specific number.
- **HOT update conditions** — refined across major versions. Accept PG12+ descriptions if the candidate names the version.